Highlights

Structurally-Calibrated Functional Attribution: A Methodology for Process Supervision Weighting in Reflective Reasoning

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- SC-FMA calibrates interventional utility into supervision weights.
- QP is strongest only on the controlled synthetic proxy benchmark.
- PRM800K real evidence is led by `w_struct`, not QP.
- Failed GSM8K/HotpotQA and downstream routes are kept out of claims.

Structurally-Calibrated Functional Attribution: A Methodology for Process Supervision Weighting in Reflective Reasoning

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ABSTRACT

Knowledge-intensive reasoning systems increasingly rely on reflective steps such as self-evaluation, error diagnosis, and plan revision to organize verification workflows. This paper presents Structurally-Calibrated Functional Attribution (SC-FMA), a methodology for auditable verification-step weighting in knowledge-intensive reasoning systems. SC-FMA converts local functional utility signals into calibrated supervision weights through the Structurally-Calibrated Utility (SCU) objective, a convex optimization problem that combines fidelity to utility, graph-level structural necessity, redundancy reduction, and bottleneck protection. On a controlled synthetic benchmark with controlled proxy step-importance labels, the full Quadratic Program (QP) variant achieves the best Spearman correlation ($\rho = 0.608$) and improves over raw CIU ($\rho = 0.483$). On PRM800K real process-supervision annotations, the primary positive evidence is instead the stored `w_struct` step-ranking signal ($\rho = 0.611$); Ridge closely preserves it ($\rho = 0.604$), while QP ($\rho = 0.442$) and Projection ($\rho = -0.135$) underperform on this real-data route. GSM8K/HotpotQA task-specific replay, downstream filtering, and PRM-training routes remain failed or unvalidated. The contribution is therefore a claim-bounded calibration methodology: SC-FMA provides formal optimization guarantees and reproducible ranking evidence, while leaving downstream deployment and external generalization to future validation.

1. Introduction

Knowledge-intensive reasoning systems require verification workflows whose intermediate checks are both locally useful and structurally well placed. In expert systems, knowledge graphs, and reflective language-agent pipelines, a verification step may correlate with final-answer success while still being redundant because another rule, entity path, or neighboring reflection supplies equivalent support. Conversely, a rare bottleneck check may carry only a weak local utility signal but still gate downstream inference. This makes process supervision a Knowledge-Based Systems (KBS) problem as well as a scoring problem: supervision weights should reflect local utility, structural dependency, redundancy, and bottleneck protection.

Existing process-supervision approaches usually treat step correctness, verifier score, or local intervention response as the supervision target [3, 8, 9]. Those signals are useful, but they do not distinguish between a locally helpful step and a structurally necessary one. Prior diagnostic FMA results show that local utility and structural necessity can be weakly aligned, so direct utility weighting can over-weight fluent but redundant reflection and under-weight structurally critical operations.

We propose **Structurally-Calibrated Functional Attribution (SC-FMA)**. SC-FMA takes local functional attribution signals and produces calibrated supervision weights by solving a constrained optimization problem over a reasoning graph. The method is designed for auditable verification-step weighting: all inputs are explicit, all calibration variants are deterministic, and claim boundaries are tied to stored artifacts.

The paper makes five contributions:

1. a convex Structurally-Calibrated Utility (SCU) objective that combines local utility, graph necessity, redundancy, and bottleneck protection;
2. three deterministic calibration variants—Ridge, Quadratic Program (QP), and Topology Projection—covering different compute budgets;

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3. formal guarantees for convexity/uniqueness, non-redundant monotonicity, variance shrinkage, and bottleneck protection;
4. controlled synthetic evidence showing QP as the strongest variant under controlled proxy labels;
5. real PRM800K evidence showing that `w_struct`, not QP, is the main real-data step-ranking result, with Ridge acting as the closest SC-FMA approximation.

This evidence boundary is intentional. The synthetic benchmark uses controlled proxy labels, not external ground truth. PRM800K supports step-label ranking and an in-distribution frozen PRM baseline comparison. GSM8K/HotpotQA task-specific replay, downstream filtering, and PRM training are not positive evidence in this submission.

2. Related Work

Process supervision and PRMs. Process Reward Models assign scalar estimates to intermediate reasoning steps, enabling verifier-guided search and process-level feedback [3, 8]. ProcessBench and related evaluations show that step-level supervision remains brittle under distribution shift [9]. SC-FMA is not a PRM-training result. It is a pre-calibration methodology that can transform existing utility or process signals into structure-aware weights before any downstream use.

Attribution for reasoning. Token-level attribution methods such as LIME, Integrated Gradients, and SHAP [4, 6, 7] explain model predictions at fine granularity, but reflective operations are multi-token functional units. SC-FMA works at the step level and calibrates functional attribution signals with graph structure.

Graph reasoning and KBS. Knowledge-graph reasoning emphasizes dependencies among entities, rules, and paths [1]. Classical KBS and cognitive-system work likewise treats inference as structured knowledge manipulation rather than isolated scoring. SC-FMA imports this structural view into reflective reasoning traces by treating verification steps as graph nodes with necessity and redundancy diagnostics.

Convex calibration. Convex optimization provides globally analyzable objectives and stable solvers [2, 5]. SC-FMA uses this tradition to make process-supervision weighting auditable: a calibrated weight is the solution to an explicit objective, not a heuristic post-processing rule.

3. Methodology

3.1. Local Utility and Intervention Boundary

Let a reasoning trace contain k reflective steps $s_1, \dots, s_k$. The local utility vector $\mathbf{c}$ records how much each step contributes under the documented intervention protocol. In the primary CIU route, interventions are length-preserving MASK replays: the targeted span is replaced with mask tokens while preserving token count and position. Trace-level binary correctness and bounded continuous utility/outcome fields in [0, 1] are supported. Fields named `correctness` remain binary by contract.

DELETE, SHUFFLE, TRUNCATE, and CONTRADICT perturbations are retained only as destructive diagnostics and robustness checks. They are useful for stress-testing structural behavior, but they are not structure-preserving primary CIU interventions. The implementation records this boundary in intervention metadata with `structure_preserving=false`, `primary_ciu_allowed=false`, and `proxy_only=true`.

3.2. Reflection Graph

SC-FMA constructs a reflection graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ whose nodes are reflective steps. Edges encode documented structural relations such as temporal adjacency and topical similarity. From this graph, the method computes:

- structural necessity $\mathbf{n}$ from graph perturbation diagnostics;
- redundancy matrix $\mathbf{R}$ from pairwise substitutability indicators;
- bottleneck indicator $\mathbf{b}$ for rare steps whose removal strongly affects downstream structure.

The graph is an operational approximation, not a claim of recovered causal structure.

Table 1
Evidence routes and claim permissions.

Route	Status	Permission
Synthetic controlled benchmark	Passed	Method calibration, ablation, and controlled proxy-label ranking.
PRM800K v3.6 frozen hash split	Passed	Real step-label ranking with `w_struct` as the primary signal.
PRM800K v3.8 frozen PRM comparison	Passed, context-only	In-distribution frozen PRM baseline context.
Stage 2 holdout	Diagnostic	Small, stratum-dependent signal only ($\rho = 0.1628$).
GSM8K/HotpotQA replay and filtering	Failed or blocked	No positive replay, filtering, PRM-training, or external-generalization claim.

3.3. SCU Objective

The calibrated weight vector $\mathbf{w}$ lies on the probability simplex. SC-FMA solves:

$$L(\mathbf{w}) = \alpha\|\mathbf{w} - \tilde{\mathbf{c}}\|_2^2 + \beta\|\mathbf{w} - \tilde{\mathbf{n}}\|_2^2 + \gamma\mathbf{w}^\top\mathbf{R}\mathbf{w} - \delta\sum_i b_i \log(w_i), \tag{1}$$

subject to $w_i \geq \epsilon$ and $\sum_i w_i = 1$. Here $\tilde{\mathbf{c}}$ and $\tilde{\mathbf{n}}$ are normalized local utility and structural necessity. The first term preserves fidelity to local utility; the second aligns weights with graph necessity; the third penalizes redundant high weights; and the fourth protects bottlenecks from collapse.

3.4. Calibration Variants

Ridge uses a closed-form softmax over a linear combination of local utility and structural necessity. It is the simplest and fastest route.

QP solves the full SCU objective with redundancy and bottleneck terms. It is the strongest synthetic-benchmark variant but is not the best PRM800K real-data variant.

Projection applies a fast topology-constrained projection. It is included as a computationally cheap ablation and is not supported as the primary PRM800K result.

Conditional replacement, counterfactual matching, and doubly robust estimation are planned extensions rather than completed evidence in this submission.

3.5. Guarantees

Under positive regularization and $w_i > \epsilon$, the SCU objective is strictly convex on the feasible simplex and has a unique global solution. For non-redundant step pairs with identical redundancy profiles, the calibrated weights preserve the joint ordering implied by $\tilde{\mathbf{c}}$ and $\tilde{\mathbf{n}}$. The structural term shrinks weight variance relative to raw utility weighting when structural necessity has lower variance than CIU, and the bottleneck log barrier prevents identified bottleneck nodes from receiving zero weight. Full derivations, KKT conditions, and solver diagnostics are provided in the supplementary material.

4. Experiments

4.1. Data and Claim Permissions

Table 1 summarizes the evidence routes used in this submission. The distinction between controlled synthetic evidence and PRM800K real-data evidence is central to the claim boundary.

4.2. Synthetic Controlled Benchmark

The synthetic benchmark contains 200 traces and 1,027 reflective steps generated by deterministic scripts. The target labels are controlled oracle/proxy step-importance labels designed for method comparison; they are not external ground truth. The oracle-independence audit in the supplementary material verifies low overlap between the proxy-label components and SCU inputs.

Table 2
Synthetic controlled step-ranking results. Mean metrics across 200 traces.

Method	Spearman ρ	Kendall τ	NDCG@3	Top-3 overlap
SC-FMA QP	**0.608**	**0.533**	**0.921**	**0.958**
SC-FMA Ridge	0.536	0.461	0.884	0.912
Monte Carlo Shapley	0.524	0.447	0.871	0.901
Gradient×Input	0.491	0.419	0.845	0.883
Raw CIU	0.483	0.411	0.839	0.875
Token surprisal	0.310	0.251	0.702	0.741

Table 3
PRM800K locked split: real step-label ranking and variant boundary.

Signal	Spearman ρ	Δ vs. `w_struct`	Interpretation
`w_struct`	**0.611**	—	Primary real-data step-ranking result.
SC-FMA Ridge	0.604	-0.007	Closest SC-FMA approximation; preserves ranking.
SC-FMA QP	0.442	-0.169	Underperforms on PRM800K; not the real-data winner.
SC-FMA Projection	-0.135	-0.746	Fails on this route.
Frozen PRM prefix score	0.252	-0.360	In-distribution baseline context only.
Raw local utility	-0.077	-0.689	Uncalibrated utility is insufficient.

On this controlled benchmark, QP is the strongest variant. It improves over raw CIU by +0.125 Spearman and the ablation study shows that structure alignment, redundancy penalty, and bottleneck protection each contribute to ranking quality. These results support the SCU optimization design under controlled conditions.

4.3. PRM800K Real Step-Label Ranking

The real-data route uses PRM800K phase-2 process-supervision annotations under a frozen hash split: 4,417 samples and 34,219 labeled steps. The primary result is `w_struct`, which reaches Spearman $\rho = 0.611$ and substantially exceeds raw local utility ($\rho = -0.077$). The frozen PRM baseline comparison is reported only as in-distribution context because it uses the same PRM800K distribution.

Two conclusions follow. First, the real-data main result is `w_struct`, not QP. Second, Ridge is the only SC-FMA variant that closely preserves the stored real-data ordering. The more complex QP and Projection variants are useful for synthetic calibration analysis but must be downgraded for PRM800K.

4.4. Stage 2 Diagnostic Holdout

The Stage 2 holdout result is intentionally reported as diagnostic. The aggregate Spearman signal is small ($\rho = 0.1628$) and stratum-dependent; high-divergence strata show stronger alignment than mid or random strata. This supports the interpretation that SC-FMA is most informative where local utility and structural necessity diverge, but it is not sufficient evidence for broad real-task validation.

5. KBS Fit and Discussion

SC-FMA is relevant to KBS research because it treats verification-step weighting as a structured knowledge problem. Instead of scoring steps independently, it asks how a step participates in a graph of dependencies, redundancies, and bottlenecks. This aligns with KBS concerns about rule interaction, knowledge-graph path redundancy, and neuro-symbolic module interfaces.

The current KBS integration evidence is methodological. A deterministic ontology-aware edge-construction pilot records ontology relation labels as diagnostic metadata and demonstrates that ontology-derived edges can change structural necessity scores. It does not validate a rule engine, ontology reasoner, KG query engine, or deployed KBS workflow. The pilot is best read as an extension path: formal ontology edges could replace keyword topical edges in future domain-specific KBS studies.

6. Scope and Limitations

The current evidence supports SC-FMA as a diagnostic and calibration methodology, not as a complete downstream training system.

- **Non-identifiability.** The intervention notation is operational shorthand for documented trace manipulation. The framework reports protocol-defined functional attribution scores, not formally identified treatment-effect parameters or global causal quantities.
- **Synthetic scope.** Synthetic results use controlled proxy labels. They justify method behavior under controlled conditions, not external truth.
- **Real-data boundary.** PRM800K supports step-label ranking and in-distribution frozen PRM baseline context. GSM8K/HotpotQA replay and filtering routes remain failed or blocked.
- **Intervention boundary.** MASK is the primary structure-preserving CIU intervention. DELETE, SHUFFLE, TRUNCATE, and CONTRADICT are destructive diagnostics only.
- **Planned components.** Conditional replacement, counterfactual matching, and doubly robust estimation remain future work and are not counted as completed evidence.
- **KBS deployment.** No production rule engine, ontology reasoner, KG query engine, or deployed KBS workflow is validated here.

7. Conclusion

SC-FMA converts local functional attribution signals into structurally calibrated supervision weights through a convex SCU objective. The controlled synthetic benchmark supports QP as the strongest method under controlled proxy labels. The PRM800K real-data route supports a different boundary: `w_struct` is the main real step-ranking signal, Ridge closely preserves it, and QP/Projection must be downgraded. With these boundaries, SC-FMA offers an auditable methodology for verification-step weighting in reflective reasoning and KBS-oriented analysis, while leaving downstream PRM training, task-specific replay validation, and deployed KBS workflows to future work.

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Declaration of Competing Interest

The authors declared that they have no conflicts of interest to this work.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the authors used OpenAI Codex for language editing, LaTeX packaging assistance, and compliance checks. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data Availability

Data will be made available on request.

CRediT authorship contribution statement

Haoran Ma: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Visualization. **Ningning Wang**: Methodology, Validation, Writing – Review & Editing, Supervision, Project administration, Funding acquisition.

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